5. Commercial & Office Buildings: Standards & Historical Benchmarks

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Primary OWL Classes: `CommercialProject`, `OfficeBuilding`, `EmpireStateBuilding`, `FamousProject`, `HistoricalProject`, `Architect`, `Steel`
Primary OWL Properties: `usesMaterial`, `usesMachine`, `hasProject`, `hasCustomer`, `year`

5.1 Ontological Structure of Commercial Projects

In `stavebna_firma_ontology - kópia.owl`, `CommercialProject` is a direct subclass of `BuildingProject`. It represents non-residential revenue-generating properties such as corporate headquarters, business centers, and retail complexes. Key ontological rules governing `CommercialProject` and its subclass `OfficeBuilding` include:

- **Architect Assignment Rule:** The class `Architect` carries the existential restriction `hasProject some CommercialProject`. In our organizational model, every licensed Architect is mandated to lead the architectural design of at least one `CommercialProject` and maintain a direct relationship with a client (`hasCustomer some Customer`).
- **Material & Machinery Requirements:** Every `CommercialProject` and `OfficeBuilding` requires `usesMaterial some Material` and `usesMachine some ConstructionMachine`.
- **Standard OfficeBuilding Temporal Restriction:** Modern catalog instances of `OfficeBuilding` have the restriction `year some integer[≥ 2000, < 2026]`, while historical benchmark subclasses can define specific historical years.

5.2 Famous Historical Benchmark: Empire State Building (1931)

Just as art ontologies include famous masterpieces to test temporal reasoning, our construction ontology includes `EmpireStateBuilding` as a canonical benchmark for structural steel high-rise engineering and temporal classification:

> **Formal OWL Axioms for `EmpireStateBuilding`:**
>
> - `rdfs:subClassOf FamousProject` (Shares the `FamousProject` superclass with `EuroveaTower`)
> - `rdfs:subClassOf OfficeBuilding` (And transitively a subclass of `CommercialProject`, `BuildingProject`, and `Project`)
> - `usesMaterial some Steel` (Constructed with a 57,000-ton structural steel frame)
> - `year value 1931` (`owl:hasValue 1931`)
>
> **Automated Temporal Inference (`HistoricalProject`):** Because `HistoricalProject` is defined via an `owl:equivalentClass` restriction as any entity with `year < 1950`, and `EmpireStateBuilding` has `year = 1931`, an OWL reasoner automatically classifies `EmpireStateBuilding` under `HistoricalProject` (a subclass of `ProjectPeriod`).

5.3 Comparison of Famous Projects in the Ontology

Attribute	`EmpireStateBuilding`	`EuroveaTower`
Superclasses	`FamousProject`, `OfficeBuilding` (`CommercialProject`)	`FamousProject`, `ApartmentBuilding` (`ResidentialProject`)
Completion Year (`year`)	`1931`	`2023`
Inferred `ProjectPeriod`	`HistoricalProject` (`year < 1950`)	`ModernProject` (`year ≥ 2000`)
Primary Ontology Material	`usesMaterial some Steel` (riveted structural steel columns & beams)	`usesMaterial some Concrete` (monolithic C60/75 reinforced concrete)
Roof / Tip Height & Floors	381 m (443.2 m to spire) | 102 floors	168 m | 46 floors

5.4 Contemporary Office Projects by ApexBuild (2020–2025)

ApexBuild's Commercial Division, headed by **Ing. arch. Richard Novák, PhD.** (`Architect`) and supervised on site by **Ing. Martin Kováč** (`SiteManager`), delivers LEED Platinum certified `OfficeBuilding` complexes across Central Europe:

- **Danube Atrium Business Center (PRJ-COM-2024-02, `year = 2024`):** 18-storey Class-A `OfficeBuilding` in Bratislava for client *Danube Business Parks* (CUST-102). Combines a composite `Steel`-and-`Concrete` structural core with a triple-glazed acoustic `Glass` curtain wall and mineral `Insulation`. Budget: EUR 42,000,000.
- **CityGate Technopark (PRJ-COM-2022-05, `year = 2022`):** 12-storey flexible IT campus utilizing long-span castellated `Steel` beams (supplied by SteelCore Europe, `Supplier`) that eliminate interior columns across 1,800 m² floor plates. Deployed two tower cranes (`Crane`) and three wheeled loaders (`Loader`).